MIT 6.1200J
Problem Set 1

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Problem 1

(a) **Theorem**: "Every natural number is tall or wide, but not both."

Answer.

We must show that for every natural number n, n is tall or wide, but not both. Let n be an arbitrary natural number. We proceed by cases.

Case 1: n is tall.
We must show n is not wide. [TODO: using the definition of tall, prove n is not wide.]

Case 2: n is not tall.
We must show n is wide and not both. [TODO: using the definition of tall, prove n is wide.] Since n is wide and not tall, n is not both tall and wide.

In both cases, n is tall or wide, but not both. Since n was arbitrary, every natural number is tall or wide, but not both. ■

(b) **Theorem**: "There exists a happy natural number between 10 and 1000 that has no happy proper divisors."

Answer.

We prove this existentially by choosing $n = 25$ as the witness.

We must verify three things:

(i) 25 is between 10 and 1000: This is clear since $10 < 25 < 1000$. ✓

(ii) 25 is happy. [TODO: using the definition of happy, verify that 25 is happy.]

(iii) 25 has no happy proper divisors. The proper divisors of 25 are 1 and 5. [TODO: using the definition of happy, verify that neither 1 nor 5 is happy.]

Therefore, $n = 25$ is a happy natural number between 10 and 1000 with no happy proper divisors, which proves the theorem. ■

(c) **Theorem**: "There are no devious natural numbers."

Answer.

We prove the equivalent statement: for every natural number n, n is not devious.

Predicate: Let $P(n)$ be the statement "n is not devious."

We proceed by induction on n, with base cases $n = 0$ and $n = 1$.

Base Case ($n = 0$): We must show $P(0)$, i.e., that 0 is not devious. [TODO: using the definition of devious, prove 0 is not devious.]

Base Case ($n = 1$): We must show $P(1)$, i.e., that 1 is not devious. [TODO: using the definition of devious, prove 1 is not devious.]

Inductive Step: Let $n \geq 1$ be an arbitrary natural number. Assume $P(n)$ holds, i.e., that n is not devious. We must show $P(n+1)$, i.e., that $n+1$ is not devious.

Assumption: n is not devious.

Goal: Show that $n+1$ is not devious.

[TODO: using the definition of devious and the inductive hypothesis that n is not devious, prove $n+1$ is not devious.]

By induction, $P(n)$ holds for all $n \in \mathbb{N}$. Therefore, there are no devious natural numbers. ■

Problem 2

(a) Brynmor's mistake is in the inductive step. When substituting the inductive hypothesis into $x_{i+1} = 6x_i - 8x_{i-1}$, he plugs in lower bounds for both x_i and x_{i-1}. However, since x_{i-1} appears with a **negative** coefficient (-8), replacing x_{i-1} with its lower bound 2^{i-1} makes the expression *smaller*, not larger. That is, from $x_{i-1} \geq 2^{i-1}$ one can only conclude $-8x_{i-1} \leq -8 \cdot 2^{i-1}$, so the inequality goes the wrong way:

$$x_{i+1} = 6x_i - 8x_{i-1} \geq 6 \cdot 2^i - 8x_{i-1}$$

is valid, but substituting the lower bound of x_{i-1} on the right gives an upper bound, not a lower bound. Hence the step

$$x_{i+1} \geq 6 \cdot 2^i - 8 \cdot 2^{i-1}$$

is **not justified**.

(b) Erik's mistake is in the base case and the circular reasoning at the end. His inductive hypothesis is $R(i) :=$ "$x_i \geq 3x_{i-1}$," defined only for $i \geq 2$. At the end, he concludes that since $x_1 > 0$ and $x_i \geq 3x_{i-1}$ for all $i \geq 2$, we get $x_i > 0$ for all i. But this final step implicitly assumes $x_{i-1} > 0$ in order to conclude $x_i \geq 3x_{i-1} > 0$, which is exactly what he is trying to prove. That is, the conclusion $x_i > 0$ requires knowing $x_{i-1} > 0$, making the argument **circular**. He never establishes by a separate argument that positivity propagates; he simply assumes it.

(c) Zach's mistake is in the inductive step when $i = 1$. He claims to prove $S(i+1)$ using $S(i)$ and $S(i-1)$, but when $i = 1$, this requires $S(0)$, i.e., $x_0 = 4^0 = 1$. However, the sequence is only defined starting from x_1; x_0 does not exist. So the inductive step **fails for** $i = 1$ because it invokes an undefined term. Indeed, the claim $S(i) :=$ "$x_i = 4^i$" is false for $i = 2$ since $x_2 = 13 \neq 16 = 4^2$, confirming the proof is broken.

(d) We prove that $x_i > 0$ for all $i \geq 1$.

Following Erik's approach, we use the stronger inductive hypothesis

$$P(i) := \text{"}x_i \geq 3x_{i-1} \text{ and } x_{i-1} > 0\text{"}$$

for $i \geq 2$, and handle $x_1 > 0$ separately. Equivalently, we prove by strong induction that $x_i > 0$ for all $i \geq 1$ and $x_i \geq 3x_{i-1}$ for all $i \geq 2$.

More precisely, let $P(i) :=$ "$x_i \geq 3x_{i-1}$" for $i \geq 2$, and we will carry the positivity of all previous terms alongside.

Claim: For all $i \geq 1$, $x_i > 0$. Moreover, for all $i \geq 2$, $x_i \geq 3x_{i-1}$.

Proof by strong induction.

Base Cases:

- $x_1 = 4 > 0$. ✓
- $x_2 = 13 > 0$, and $x_2 = 13 \geq 12 = 3 \cdot 4 = 3x_1$. ✓

Inductive Step: Let $i \geq 2$. Assume that for all $1 \leq k \leq i$, $x_k > 0$, and for all $2 \leq k \leq i$, $x_k \geq 3x_{k-1}$. We wish to show that $x_{i+1} > 0$ and $x_{i+1} \geq 3x_i$.

Consider:

$$x_{i+1} = 6x_i - 8x_{i-1}.$$

By the inductive hypothesis, $x_i \geq 3x_{i-1}$, so $x_{i-1} \leq \frac{x_i}{3}$, and thus $8x_{i-1} \leq \frac{8}{3}x_i$. Therefore:

$$x_{i+1} = 6x_i - 8x_{i-1} \geq 6x_i - \frac{8}{3}x_i = \frac{10}{3}x_i.$$

Since $x_i > 0$ by the inductive hypothesis, we have:

$$x_{i+1} \geq \frac{10}{3}x_i > 3x_i > 0.$$

This establishes both $x_{i+1} \geq 3x_i$ and $x_{i+1} > 0$, completing the inductive step.

By strong induction, $x_i > 0$ for all $i \geq 1$. ■